Aryan Raval

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Education

South Dakota School of Mines & Technology, Bachelor of Science in Computer Science — Aug 2022 – May 2026

- GPA: 3.925/4.0
- Outstanding Computer Science Junior Award at the 74th Honors Convocation

Experience

C-lock Inc., Software Engineer — Dec 2025 – Nov 2026

Rapid City, South Dakota

- Maintained an R-based anomaly detection script monitoring 10,000+ IoT devices, applying statistical methods to gas, proximity, RFID, and load-cell sensor data to flag drift and hardware failures for daily engineering review.
- Optimized baseline algorithm that tracks feed consumption across 1,000+ IoT devices using rolling medians and variance masking to filter animal induced weight fluctuations and track refill events.
- Built a PHP experiment scheduling platform serving 1,000+ customers; migrated legacy Salesforce records and integrated RBAC REST APIs, ClickUp automation, and vision LLM image analysis.
- Secured PHP REST APIs with IP-based rate limiting, brute-force lockouts, and standardized JSON responses; built an MCP server integrating these APIs for AI-driven device queries and diagnostics.

Caterpillar, Software Development Intern — Jan 2025 – May 2025

South Dakota Mines

- Enhanced a Next.js full-stack platform on Azure used by Caterpillar employees worldwide; implemented RBAC via Azure AD and integrated internal databases through REST APIs secured with OAuth 2.0 client credentials.
- Managed Azure development, QA, and production environments, configuring secure database connectivity and secret management with Azure Managed Identity and Key Vault.
- Coordinated weekly with stakeholders to deliver features, modernized UI/UX, moved the application to a public facing domain, and integrated SendGrid for automated notifications.

South Dakota Mines, Undergraduate Research Intern — Nov 2024 – Nov 2025

- Researched on Transmission Electron Microscopy (TEM) image analysis in Python using OpenCV and K means clustering to denoise grayscale micrographs, segment nanoparticles from the background, and automate particle sampling.
- Built an interactive 3D Atomic Force Microscope (AFM) educational platform using React, Three.js, and Blender, with component exploration, animated nanoscale scanning, and simulated image formation (nanoafm.netlify.app).

Projects

Machine Learning Fraud Detection | GitHub | *Python, Pandas, NumPy, scikit-learn, XGBoost*

- Built a Python ETL pipeline for 590K labeled online payment transactions from Kaggle, joining transaction and identity records to prepare financial data for detecting fraudulent purchases.
- Trained and compared Logistic Regression, Random Forest, and XGBoost for fraud classification, with XGBoost achieving the strongest test performance across precision, recall, F1, and ROC-AUC.
- Developed a command line pipeline with chronological data splits, validation based threshold selection, and automated saving of trained models, transaction predictions, and evaluation reports.

Degree Planner – Senior Design | *React, TypeScript, Express, MongoDB, Azure AD*

- Built an interactive degree-planning platform using React Flow, with reactive validation of prerequisites, corequisites, and semester course-offering constraints.
- Developed Express/MongoDB REST APIs for degree plans and curriculum management, implementing role-based access control and Azure AD OAuth 2.0 authentication.
- Deployed the full-stack application on a Linux server using Nginx as an HTTPS reverse proxy and PM2 for process management.

Skills

Languages: C++, Python, SQL, JavaScript, TypeScript, PHP, R, C#, ARM Assembly

Frameworks and Libraries: React.js, ASP.NET, Django, Next.js, Tailwind, Pandas, OpenCV, sciki-learn

Tools & Technologies: Azure, Docker, Node.js, Nginx, GitLab, GitHub, SSMS, Firebase, Blender, PowerBI